

Student ID: 202312097

IT615 – Data Base Management System
Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)
First In-Sem Examination, September 2023

[Time – 2 Hours]

[Total Marks - 60]

Instructions:

- Answer all question. All questions are self-explanatory and understanding of question is a part of evaluation.
- No query regarding questions will be entertained during examination by course instructor or invigilator.
- Space for Rough work is given at last two pages (Page No. 13 and 14). You need to do rough work only at that place.

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Student Name:

THADANI VISHAL GOVIND

Marks Obtained		
Question No.	Maximum Marks	Obtained Marks
1	3	2
2	2	2
3	2	2
4	2	1
5	1	1
6	1	1
7	1	1
8	4	0
9	4	4
10	2	2
11	20	15
12	8	6
13	10	8
Total Marks	60	43 + 2 = 45

1. Consider the bank database as given below.

[3 Marks]

branch(branch name, branch city, assets)
 customer (ID, customer name, customer street, customer city)
 loan (loan number, branch name, amount)
 borrower (ID, loan number)
 account (account number, branch name, balance)
 depositor (ID, account number)

Assume that branch names and customer names uniquely identify branches and customers, but loans and accounts can be associated with more than one customer.

- What are the appropriate primary keys?
- Given your choice of primary keys, identify appropriate foreign keys.

primary keys

a) branch → branch name

(b)

c customer → ID

loan → loan number

borrower → ID

account → account number

depositor → ID

2

2. What are integrity constraints? Why they are important?

[2 Marks]

→ Integrity constraints are constraints which preserves the integrity of values of attributes across different tables.

→ We use primary key in one table as foreign key in another,

2 So that foreign key value will only belong to set of values in the table it is primary key.

3. What is an instance? What is a schema?

[2 Marks]

→ Instance is the data's ~~actual~~ view at a point of time when it has values

→ Schemas are views to represent data.

Physical → ~~map~~ memory & allocated

logical → ~~map~~ conceptual model

②

4. Explain the concept of physical data independence and its importance in database systems?

[2 Marks]

→ Data is ~~independent~~ from changes made in logical model.

→ Physical memory change has no effect

①

5. SELECT operation in SQL is equivalent to

- (1). The Selection operation in relational algebra
- (2). The Selection operation in relational algebra, except that SELECT in SQL retains duplicates
- (3). The Projection operation in relational algebra
- ✓ (4). The Projection operation in relational algebra, except that SELECT in SQL retains duplicates

[1 Marks]

6. Which of the following statements are TRUE about an SQL query? Tick in below options

- P: An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause
- Q: An SQL query can contain a HAVING clause only if it has a GROUP BY clause
- R: All attributes used in the GROUP BY clause must appear in the SELECT clause
- S: Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

(1) P and R

(2) P and S

✓ (3) Q and R

(4) Q and S

[1 Marks]

7. Rewrite the query

select *

from section natural join classroom

without using a natural join but instead using an inner join with a using condition.

SELECT *
FROM section INNER JOIN classroom
USING class_id

→ (NOTE: Since the common attribute between section & classroom tables, are not specified in question, assume every section has class_id named attributed which references class_id from classroom)

[1 Marks]

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8. Consider the relational database with the following four schemas and their respective instances.

Student(sNo, sName, dNo)

Dept (dNo, dName)

Course(cNo, cName, dNo)

Register(sNo, cNo)

Student		
sNo	sName	dNo
S01	James	D01
S02	Rockey	D02
S03	Jacky	D02
S04	Jini	D01
S05	Mili	D02

Dept	
dNo	dName
D01	CSE
D02	EEE

Course		
cNo	sName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

SQL Query :

SELECT * FROM Student AS S WHERE NOT EXIST

(SELECT cNo FROM Course WHERE dNo = "D01"

EXCEPT

SELECT cNo FROM Register WHERE sNo = S.sNo)

The number of rows returned by the above SQL query is 4.

[4 Marks]

The above query returns all details of all students who have registered in all courses from 'D01' department.

We have 2 such students, S01, S02 but since SELECT DISTINCT IS NOT USED we get 4 rows.

9. Consider the set of relations

EMP (Employee-no, Dept-no, Employee-name, Salary)

DEPT (Dept-no, Dept-name, Location)

Write an SQL query to:

1. Find all employee names who work in departments located at "Calcutta" and whose salary is greater than Rs.50,000.
2. Calculate, for each department number, the number of employees with a salary greater than Rs.1,00,000.

[4 Marks]

1. SELECT Employee-name
FROM EMP NATURAL JOIN DEPT
WHERE Location = 'Calcutta'
AND Salary > 50000



2. SELECT COUNT(*), dept-no
FROM EMP
GROUP BY dept-no
HAVING salary > 100000



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10. Consider the following relations

- **Department** (dept_name, building, budget,
PRIMARY KEY (dept_name));
- **Course** (course_id, title, dept_name, credits,
PRIMARY KEY (course_id),
FOREIGN KEY (dept_name) **REFERENCES** department);
- **SECTION** (course_id, sec_id, semester, year, building, room_number, time_slot_id, **PRIMARY KEY** (course_id, sec_id, semester, year),
FOREIGN KEY (course_id) **REFERENCES** course);

Consider the SQL statement below:

```
SELECT T.course_id
FROM course AS T
WHERE unique
      (SELECT R.course_id
       FROM section AS R
       WHERE T.course_id = R.course_id
       AND R.year = 2018);
```

Which of the following is the correct interpretation for this SQL query?

- (a) Find all courses that were offered till in 2018
- ☒ (b) Find all courses that were offered at most once in 2018
- (c) Find all courses that were offered more than once in 2018
- (d) Find all courses offered in 2018 multiple times

[2 Marks]

11. Consider the insurance database as given below. Primary keys are underlined.

person (driver id, name, address)
car (license plate, model, year)
accident (report number, year, location)
owns (driver id, license plate)
participated (report number, license plate, driver id, damage amount)

Construct the following SQL queries for this relational database.

1. Find the total number of people who owned cars that were involved in accidents in 2017.
2. Delete all year-2010 cars belonging to the person whose ID is '12345'.
3. Find the number of accidents involving a car belonging to a person named "John Smith".
4. Update the damage amount for the car with license plate "AABB2000" in the accident with report number "AR2197" to \$3000.

[8 Marks]

1. SELECT COUNT(*) AS total-people
FROM (SELECT DISTINCT owns.driver_id
FROM accident, owns, participated

```
WHERE accident.report-number = participated.report-number
AND participated.license-plate = owns.license-plate
AND accident.year = 2017)
```

2. DELETE car
FROM (SELECT *
FROM car NATURAL JOIN owns
WHERE driver-id = '12345'
AND year = 2010)

3. SELECT COUNT(*) AS no-of-accidents
FROM (SELECT report-number
FROM owns NATURAL JOIN participated
USING license-plate, person
WHERE ~~owns.person-id =~~
owns.driver-id = person.driver-id
AND person-name = 'John Smith')

2

4. UPDATE ~~damage-amount~~ participated
FROM (SELECT *
FROM participated
WHERE license-plate = 'AABB2000'
AND report-number = 'AR2197')
SET damage-amount = 3000

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12. Consider the library database as given below. Primary keys are underlined

member (memb no, name)

book (isbn, title, authors, publisher)

borrowed (memb no, isbn, date)

Write the following SQL queries

1. Find the member number and name of each member who has borrowed at least one book published by "McGraw-Hill".
2. Find the member number and name of each member who has borrowed every book published by "McGraw-Hill".
3. For each publisher, find the member number and name of each member who has borrowed more than five books of that publisher.
4. Find the average number of books borrowed per member. Take into account that if a member does not borrow any books, then that member does not appear in the *borrowed* relation at all, but that member still counts in the average.

[10 Marks]

1)
 SELECT ^{DISTINCT} memb_no, name
 FROM member NATURAL JOIN borrowed, book
 WHERE borrowed.isbn = book.isbn
 AND publisher = 'McGraw-Hill'

2) SELECT DISTINCT A.member-number, A.name
 FROM member AS A NATURAL JOIN borrowed, book
 WHERE borrowed.isbn = ~~book.isbn~~ AND

NOT EXIST

(SELECT isbn FROM book WHERE publisher = 'McGraw-Hill'

EXCEPT

SELECT isbn FROM borrowed^{AS} B WHERE B.memb-no =
A.memb-no)

2 ✓

3. SELECT DISTINCT memb-no, name

FROM (SELECT count(*) AS no-of-books, memb-no, publisher, name

FROM member NATURAL JOIN borrowed, book

WHERE borrowed.isbn = book.isbn

GROUP BY memb-no, publisher)

2 ✓

WHERE no-of-books > 5

4.

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13. Consider the following information about a university database:

- Professors have an SSN, a name, an age, a rank, and a research specialty.
- Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget.
- Graduate students have an SSN, a name, an age, and a degree program (e.g., M.S. or Ph.D.).
- Each project is managed by one professor (known as the project's principal investigator).
- Each project is worked on by one or more professors (known as the project's co-investigators).
- Professors can manage and/or work on multiple projects.
- Each project is worked on by one or more graduate students (known as the project's research assistants).
- When graduate students work on a project, a professor must supervise their work on the project. Graduate students can work on multiple projects, in which case they will have a (potentially different) supervisor for each one.
- Departments have a department number, a department name, and a main office.
- Departments have a professor (known as the chairman) who runs the department.
- Professors work in one or more departments, and for each department that they work in, a time percentage is associated with their job.
- Graduate students have one major department in which they are working on their degree.

Design and draw an ER diagram that captures the information about the university. Use only the basic ER model here; that is, entities, relationships, and attributes. Be sure to indicate any key and participation constraints. Indicate primary key, foreign key (if any) in each entities.

[20 Marks]

primary key is indicated by underline on page behind.

Constraint is SSN should be unique all over.

